These are the main components shown in figure 7.2:

- *Cluster*—A set of nodes running containerized applications. It hosts the Control Plane and comprises one or more worker nodes.
- *Control Plane*—The cluster component exposing the API and interfaces to define, deploy, and manage the life cycle of Pods. It comprises all the essential elements that implement the typical features of an orchestrator, like cluster management, scheduling, and health monitoring.
- *Worker nodes*—Physical or virtual machines providing capacity such as CPU, memory, network, and storage so that containers can run and connect to a network.
- *Pod*—The smallest deployable unit wrapping an application container.

Now that you have a good understanding of the Kubernetes infrastructure, let's see how you can create and manage a Kubernetes cluster on your local machine.

7.1.1 Working with a local Kubernetes cluster

In chapter 2, we worked with *minikube* (https://minikube.sigs.k8s.io), a tool for running Kubernetes clusters on local environments. We used the minikube CLI to create a local Kubernetes cluster relying on the default configuration. In this section, you'll see how to define a custom configuration for minikube that you can use when initializing a new local Kubernetes cluster for deploying Polar Bookshop.

NOTE If you haven't installed minikube yet, refer to the instructions in section A.3 of appendix A.